Highlights

Two-scale stochastic simulation of epidemics on community networks with aggregate inter-community coupling

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- Exact local Gillespie engines are coupled through aggregate inter-community hazards
- The ideal scheme is exact simulation of a surrogate all-to-all interface network
- Aggregate export clocks are stochastically early for randomly placed interfaces
- Accuracy is organised by internal timing spread relative to edge crossing time
- Close arrival times can coexist with physically impossible transmission routes

Two-scale stochastic simulation of epidemics on community networks with aggregate inter-community coupling

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Abstract

Multiscale stochastic simulation often couples detailed subsystem dynamics through aggregate interfaces. We study this problem for epidemic spreading on community networks, where within-community contact structure is represented explicitly while inter-community coupling is available only in aggregate form. Each community is evolved by a Gillespie engine that is exact conditional on the imported events it receives; the macro layer generates imports from aggregate, state-dependent hazards and, in the default implementation, places them uniformly among susceptible nodes of the receiving community. We give a complete specification of the method, including its grid-based synchronisation, and show that its ideal continuous-time version is an exact simulation of a surrogate network in which each bundle of inter-community edges is replaced by uniform all-to-all coupling of equal total weight. This representation identifies four sources of discrepancy: loss of boundary-state information, uniform import placement, absence of interface-edge exhaustion, and synchronisation error. For uniformly random interfaces in the single-export setting, we further show that the ideal aggregate export clock is stochastically earlier than the full-network one. Controlled experiments on synthetic SI networks compare the method with full-network Gillespie simulation in single-export benchmarks and in 48 configurations of ten-community chains (50–1000 nodes per community, internal path lengths 1–6, one or five interface edges, 400 matched runs each). When the spread of infection times within a community was below a quarter of the mean single-edge crossing time, terminal arrival-time distributions were statistically indistinguishable in 15 of 18 configurations; beyond this regime, the two-scale simulator arrived early by an amount roughly proportional to that ratio, reaching 40%. Close timing did not imply admissible routes: up to 52% of aggregate crossings preceded infection of any true bridge node.

1. Introduction

Stochastic epidemic simulation in structured populations must account for dynamics occurring at different spatial and structural scales. Within communities, transmission can depend on detailed contact networks and local stochastic interactions (Keeling and Eames, 2005; Pastor-Satorras et al., 2015; Kiss et al., 2017), whereas transmission between communities is often represented through aggregate mobility flows or weighted coupling networks (Balcan et al., 2009; Barbosa et al., 2018). This mismatch in resolution creates a multiscale simulation problem in which fine-grained local dynamics must be coupled through a coarser inter-community representation.

In practice, such a mismatch is common because within-community contact networks may be observed, reconstructed or synthesised from partial information (Eubank et al., 2004; Danon et al., 2011), whereas interactions between communities may be available only as commuting volumes, mobility counts or other aggregate coupling measures (Barbosa et al., 2018; Chang et al., 2021). Importantly, an aggregate measure of connectivity between two communities does not uniquely determine the microscopic cross-community contacts that produced it. Consequently, the aggregate representation omits information about which

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individuals constitute the interface between communities and how their states affect inter-community transmission.

When the complete contact network is available, epidemic dynamics can be formulated as a continuous-time Markov chain (CTMC) and sampled exactly by Gillespie-type event-driven methods (Gillespie, 1977; Gibson and Bruck, 2000). Extensions for temporal, large and sparse networks can substantially improve computational efficiency (Vestergaard and Génois, 2015; Kuryliak et al., 2025), but they still require the relevant microscopic interaction channels to be explicitly represented. At the opposite end of the modelling spectrum, metapopulation approaches naturally describe interactions between communities through aggregate flows, but usually aggregate the within-community state as well (Colizza et al., 2007; Colizza and Vespignani, 2008; Balcan et al., 2010; Ajelli et al., 2010). This leaves an intermediate simulation problem: how to retain explicit stochastic dynamics within communities while coupling them through an inter-community interface that is available only in aggregate form.

A two-scale simulator for this intermediate regime was outlined in a short workshop paper (Kuryliak and Emmerich, 2025). Each community is evolved by an independent stochastic micro-simulator, while a macro layer generates cross-community imports from state-dependent aggregate hazards and places each import in the receiving community. This architecture preserves detailed stochastic dynamics within communities while aggregating only the coupling interface. The workshop paper motivated the architecture and sketched its layers, but did not quantify the fidelity loss caused by replacing microscopic cross-community channels by aggregate coupling or by the rule used to place imported events. We therefore present the method here in a self-contained, implementation-level form.

The central question of this article is whether an aggregate inter-community interface can reproduce the stochastic behaviour of the corresponding full-network CTMC with sufficient fidelity, for which observables and in which network regimes. The answer depends on information discarded by the interface: which nodes carry the inter-community channels and whether they are currently infected or susceptible, where imported infections are placed, and, for thin interfaces, how many distinct microscopic crossings are physically available. We address these effects analytically and through controlled experiments on synthetic networks in which the complete microscopic network is known to the experimenter but only its aggregate representation is passed to the two-scale simulator. The purpose is methodological rather than empirical calibration of a particular epidemic system.

The contributions are as follows.

1. A complete, implementation-level specification of the two-scale method, including the grid-synchronised hazard-integral scheduler used in practice (Section 3, Algorithm 1).
2. An exact characterisation of the ideal method as full-network simulation on a surrogate network with uniform all-to-all interface coupling (Proposition 1). It yields a taxonomy of discrepancies and a stochastic-ordering result for single exports (Proposition 2).
3. A controlled comparison with full-network Gillespie simulation on single-export benchmarks and on chains of communities, using permutation baselines that separate approximation error from sampling variability (Sections 5 and 6).
4. A practical criterion: the ratio of the within-community spread of infection times to the single-edge crossing time organises the observed discrepancies across community sizes, internal structures and interface widths (Section 6.6).

The study is restricted to susceptible–infected (SI) dynamics on static synthetic networks and a fixed synchronisation step. Section 2 relates the method to existing simulation approaches; Section 3 specifies it; Section 4 analyses what the aggregate interface preserves; Sections 5 and 6 describe the experiments and their results; Section 7 discusses practical use, cost and limitations.

2. Related work

Exact and approximate stochastic simulation. Gillespie's direct method samples a CTMC exactly from the current propensities (Gillespie, 1977); the next reaction method organises the same computation around per-channel clocks (Gibson and Bruck, 2000). When propensities vary in time, exact simulation can be expressed through the random-time-change representation of a counting process (Ethier and Kurtz, 1986):

an event fires when its integrated intensity reaches an independent exponential threshold. Anderson (2007) used this representation to derive the modified next reaction method for time-dependent propensities and delays. The macro clock of the two-scale simulator uses the same cumulative-hazard principle, but with an intensity driven by the evolving stochastic states of the local engines. Network-oriented implementations update only the propensities affected by an event (Vestergaard and Génois, 2015; Kuryliak et al., 2025); the local engines used here follow Kuryliak et al. (2025).

Hybrid and multiscale stochastic simulation algorithms accelerate systems with widely separated reaction rates by approximating fast channels (Haseltine and Rawlings, 2002; Cao et al., 2005). The two-scale method differs in what it approximates: it does not average fast reaction channels, but removes microscopic information about which inter-community channels are active. As shown later, its accuracy is nevertheless organised by a separation of time scales between internal spread and interface crossing.

Metapopulation and mobility models. Metapopulation models couple subpopulations through mobility flows and have been used to study invasion thresholds and global spread (Colizza et al., 2007; Colizza and Vespignani, 2008; Balcan et al., 2009, 2010); high-resolution mobility networks can likewise be combined with local transmission models (Chang et al., 2021). Agent-based models resolve individuals throughout (Eubank et al., 2004; Kerr et al., 2021), and Ajelli et al. (2010) compared the two paradigms. In contrast, the present method preserves explicit node-level stochastic dynamics within communities and applies coarse-graining only at the inter-community interface, where microscopic transmission channels are replaced by aggregate stochastic coupling.

Two levels of mixing and co-simulation. Epidemic models with two levels of mixing combine local contacts within small groups with uniform global mixing (Ball et al., 1997). The surrogate network of Proposition 1 has a related structure, with uniform mixing restricted to pairs of communities and weighted by their aggregate coupling. From a simulation-engineering perspective, the two-scale scheduler is also related to co-simulation: independently solved subsystems are advanced on a communication grid and exchange a limited set of coupling variables (Gomes et al., 2018). The distinction is that the interface here generates stochastic event times and target locations rather than merely exchanging state variables, and therefore directly determines part of the simulated event process.

3. The two-scale simulation method

This section defines the full-network microscopic reference process and the two-scale simulator used throughout the study. The reference model is a CTMC on the specified contact network. The two-scale simulator preserves its within-community transition channels but replaces explicit cross-community channels by an aggregate stochastic interface. We distinguish the ideal continuous-time two-scale process from the grid-synchronised implementation used in the experiments.

3.1. Full-network reference model

Let $G = (V, E)$ be a contact network whose node set is partitioned into M disjoint communities,

$$V = \bigsqcup_{i=1}^{M} V_i, \qquad N_i = |V_i|. \tag{1}$$

We treat E as a set of oriented transmission channels; an undirected contact contributes two channels. The channels are divided into within-community bundles E_{ii} and inter-community bundles

$$E_{ij} = \{(u, v) \in E : u \in V_i,\, v \in V_j\}, \qquad i \neq j.$$

Each node is susceptible (S), infected (I) or removed (R). The microscopic state is $X(t) \in \{S, I, R\}^V$, with $X_i(t) = X(t)|_{V_i}$ denoting its restriction to community V_i. We write

$$S_i(t) = \{v \in V_i : X_v(t) = S\}, \qquad I_i(t) = \{v \in V_i : X_v(t) = I\},$$

Item	Full	TwoScale
Within-community contacts and rates	yes	yes
Node states within communities	yes	yes (local engines)
Inter-community channels E_{ij} (bridge nodes and endpoints)	yes	no
Aggregate weight W_{ij}	implied by E_{ij}	yes
Rate of transmission from V_i to V_j	from states of channels in E_{ij}	from $\bar{y}_i$, $\bar{s}_j$ and W_{ij}
Receiving node of a transmission	endpoint of the firing channel	drawn from a placement rule
Maximum transmissions across a bundle (SI)	number of channels	limited only by s_j

Table 1: Information used by the full-network reference and by the two-scale simulator.

and use $s_i(t) = |S_i(t)|$ and $y_i(t) = |I_i(t)|$ for the corresponding susceptible and infected counts. Their normalized fractions are $\bar{s}_i(t) = s_i(t)/N_i$ and $\bar{y}_i(t) = y_i(t)/N_i$.

For a configuration $x \in \{S, I, R\}^V$, let $x^{v:I}$ and $x^{u:R}$ denote the configurations in which node v has become infected or node u has been removed. The full-network process is the CTMC with generator

$$(\mathcal{L}_G f)(x) = \sum_{(u,v)\in E} \beta_{uv}\, \mathbf{1}\{x_u = I,\, x_v = S\}\big[f(x^{v:I}) - f(x)\big] + \sum_{u\in V} \gamma\, \mathbf{1}\{x_u = I\}\big[f(x^{u:R}) - f(x)\big], \qquad (2)$$

for bounded test functions f, where β_{uv} is the transmission rate of channel (u, v) and γ the removal rate. Inter-community rates are written as $\beta_{uv} = \beta_{\text{out}} w_{uv}$ with contact weights w_{uv}. We call exact simulation of (2) Full. All experiments in this article use SI dynamics ($\gamma = 0$).

3.2. Aggregate interface

The two-scale simulator, denoted TwoScale, receives the within-community graphs $G_i = (V_i, E_{ii})$ with their rates, but not the microscopic inter-community bundles. Each bundle is represented only by its aggregate weight

$$W_{ij} = \sum_{(u,v)\in E_{ij}} w_{uv}, \qquad i \neq j, \qquad (3)$$

which for unit weights is the number of channels from V_i to V_j. In applications, W_{ij} may be estimated from mobility or contact data rather than computed from a known microscopic network. The macro layer observes the community summaries (s_i, y_i), or equivalently $(\bar{s}_i, \bar{y}_i)$, but not the identities and states of the nodes incident to the inter-community channels. Table 1 lists the information used by each model, and Figure 1(a,b) contrasts the two representations.

3.3. Local engines

Each community is simulated by a separate engine that samples the CTMC of its own graph G_i: infection along channels in E_{ii} and removal of infected nodes, with the propensities of (2) restricted to V_i. The engines use the incremental Gillespie implementation of Kuryliak et al. (2025), in which each node stores the weight of its contacts with infected neighbours and only the propensities affected by an event are updated. Each engine owns a private pseudorandom stream.

An engine supports two operations used by the macro layer: advancing its clock to a prescribed time, applying only events that occur before it, and an *import*, which sets a given susceptible node to I. An import changes the local state and propensities, so the engine discards its pending next event and samples a new one from the updated state. This restart is distributionally valid for the CTMC because the relevant waiting times are exponential. Equivalently, a random-time-change implementation may retain the unconsumed exponential hazard budget across the import. Conditional on the complete sequence of import times and target nodes, the local engine therefore has exactly the law of the within-community CTMC with those

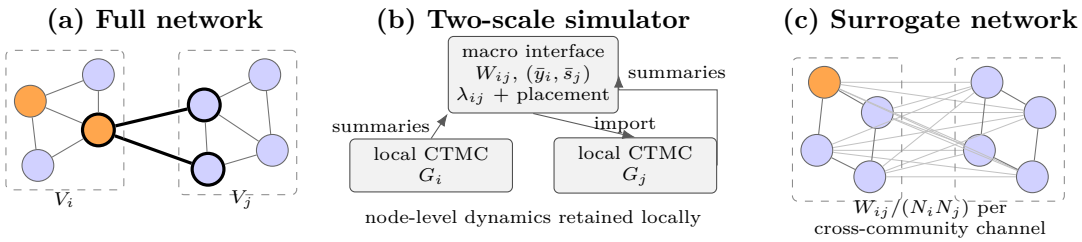


Figure 1: Conceptual comparison of the inter-community representation. (a) In the full network, infection can cross communities only along explicit channels incident to bridge nodes (thick black rings). (b) In the two-scale simulator, each community remains a local Gillespie/CTMC engine, while the macro interface uses community summaries and the aggregate weight W_{ij} to generate an import rate and target placement. (c) Under uniform placement, the ideal two-scale process is equivalent to the surrogate network of Proposition 1, in which every ordered cross-community node pair receives the weak rate $\beta_{\text{out}} W_{ij}/(N_i N_j)$. Orange denotes infected nodes and blue susceptible nodes.

imports applied as exogenous $S \to I$ jumps. The approximation enters through how the import process is generated, not through the local CTMC simulation itself.

3.4. Macro layer

The macro layer treats communities as nodes of a weighted graph with weights W_{ij}. Given the current summaries, it assigns to every ordered pair $i \neq j$ the hazard

$$\lambda_{ij}(t) = \beta_{\text{out}}\, W_{ij}\, \bar{y}_i(t)\, \bar{s}_j(t), \qquad \Theta(t) = \sum_{i \neq j} \lambda_{ij}(t). \tag{4}$$

A macro event represents one imported infection from some community into another. Conditional on a macro event at time t with $\Theta(t) > 0$, the pair (i, j) is selected with probability $\lambda_{ij}(t)/\Theta(t)$, and a target $v \in S_j(t)$ is drawn from a placement rule $\Pi_{ij}(\cdot \mid X(t))$. The default rule used throughout this article is uniform placement, $\Pi_{ij}(v \mid X(t)) = 1/s_j(t)$. Degree-weighted or boundary-aware rules are possible when such information is available. The implementation also contains a multiplicative calibration factor for (4), which is set to one in all experiments.

Equation (4) is an effective aggregate intensity rather than, in general, the exact rate of the realised microscopic bundle. Conditional on the community counts, it equals the expected bundle rate when the inter-community endpoints are assigned uniformly at random, before conditioning on their realised identities. It coincides with the actual rate when the interface is uniform all-to-all coupling (Corollary 1) and is otherwise an approximation (Section 4). The macro layer is not autonomous: Θ changes whenever a local event or import changes a community count. Because the summaries are piecewise constant between such events, so is Θ.

3.5. Hazard-integral timing and synchronisation

Let t_0 be the time of the previous macro event and $E \sim \text{Exp}(1)$ an independent threshold. In the ideal continuous-time construction, the next macro event occurs at

$$t^* = \inf\Big\{ t > t_0 : \int_{t_0}^{t} \Theta(s)\, ds \geq E \Big\}. \tag{5}$$

Conditional on the realised micro-layer trajectories, $\Theta(t)$ is a predictable intensity path and (5) is the usual random-time-change or cumulative-hazard construction (Ethier and Kurtz, 1986; Anderson, 2007). Unconditionally, the macro-event process has a stochastic predictable intensity rather than the deterministic intensity of an ordinary non-homogeneous Poisson process. For constant Θ, (5) reduces to Gillespie's waiting time $(t^* - t_0) = -\ln U/\Theta$, $U \sim \mathcal{U}(0,1)$. Combined with exact local engines, this construction defines the *ideal* two-scale process, whose transition rates are characterised in Section 4.1.

Algorithm 1 Grid-synchronised two-scale simulation of one realisation.

Input: local graphs G_i, weights W_{ij}, rates, step τ, horizon T_{end}, initial state.

$t \leftarrow 0$; $A \leftarrow 0$; draw $E \sim \text{Exp}(1)$
while $t < T_{\text{end}}$ and infection can still spread **do**
  $t_g \leftarrow \min(t + \tau, T_{\text{end}})$
  while $t < t_g$ **do**
    $\Delta \leftarrow t_g - t$; copy every engine with its random-number state; advance the copies to $t + \Delta/2$
    $\Theta_{\text{mid}} \leftarrow \sum_{i \neq j} \lambda_{ij}$ evaluated on the copies, Eq. (4)
    if $A + \Theta_{\text{mid}}\Delta < E$ **then** // no macro event before t_g
      advance all engines to t_g; $A \leftarrow A + \Theta_{\text{mid}}\Delta$; $t \leftarrow t_g$
    else
      $t^* \leftarrow t + (E - A)/\Theta_{\text{mid}}$; advance all engines to t^*
      sample (i, j) with probability $\lambda_{ij}(t^*)/\Theta(t^*)$ from the engines' states
      draw v uniformly from $S_j(t^*)$; import v into engine j (its pending event is discarded)
      $t \leftarrow t^*$; $A \leftarrow 0$; draw a new $E \sim \text{Exp}(1)$

The implemented scheduler (Algorithm 1) treats the local engines as black boxes that expose their state only when they are advanced. It divides time into synchronisation intervals of length τ. Let A denote the approximate cumulative macro hazard accumulated since the previous macro event. On each (sub)interval $[t, t_g)$, with $\Delta = t_g - t$, the scheduler estimates Θ by a midpoint probe: copies of all engines, including their pseudorandom-number states, are advanced to $t + \Delta/2$, and Θ_{mid} is computed from those copies. If $A + \Theta_{\text{mid}}\Delta < E$, the actual engines are advanced to t_g. Otherwise, the crossing time is interpolated as

$$t^* = t + \frac{E - A}{\Theta_{\text{mid}}}, \tag{6}$$

the engines are advanced to t^*, and the pair and target are sampled from their states at t^*. The threshold is then renewed and the remainder of the interval is processed in the same way.

Two properties of the scheduler matter for interpreting the results. First, because the midpoint probes copy the pseudorandom-number states, a probe reproduces the trajectory that the corresponding engine will actually follow up to the midpoint unless an import intervenes. The midpoint rule is therefore a quadrature of the realised piecewise-constant hazard. It is exact on subintervals containing no local event in any community; its quadrature error and its look-ahead of at most $\tau/2$ vanish as $\tau \to 0$. Second, if local engines expose their next-event times, the black-box restriction is unnecessary. Since Θ is piecewise constant between local events, integral (5) can then be evaluated exactly by an event-driven scheduler that merges the next-event times of all engines. This removes the numerical synchronisation approximation. We evaluate the grid-based variant, which is the implemented method, with $\tau = 1$.

3.6. Computational structure and intended use

Full stores every channel, including all inter-community channels, and its cost per event grows with node degree (Kuryliak et al., 2025). TwoScale stores the within-community graphs and an $M \times M$ weight matrix. This is the saving emphasised in the workshop paper (Kuryliak and Emmerich, 2025): when inter-community contacts are numerous, for example when a mobility flow corresponds to many individual contacts, the interface reduces to $O(M^2)$ numbers. Local engines can also be modified independently, which makes community-specific interventions straightforward to represent. Against this, the grid-synchronised scheduler adds a cost proportional to $(T_{\text{end}}/\tau)\sum_i N_i$ for copying engine states, independent of the number of events. The method is therefore suited to settings in which the inter-community contacts are unknown or too numerous to represent, rather than to small networks whose full contact structure is available. Measured run times are reported in Section 7.2.

4. What the aggregate interface preserves and what it loses

4.1. Surrogate-network representation

With exact timing (5), the ideal two-scale process is a CTMC on $\{S, I, R\}^V$. Its rate of the transition $x \to x^{v:I}$ by an import is $\sum_{i\neq j} \lambda_{ij}(x)\Pi_{ij}(v \mid x)$ for $v \in S_j(x)$; all local transitions have the rates of (2) restricted to E_{ii}. The following proposition identifies this process with a full-network process on a modified graph.

Proposition 1 (Surrogate network). *Let $\tilde{G}$ be the network with the same within-community channels and rates as G, in which every inter-community bundle E_{ij} is replaced by the complete set of channels $V_i \times V_j$, each with rate $\tilde{\beta}_{uv} = \beta_{\mathrm{out}} W_{ij}/(N_i N_j)$. Under uniform placement, the ideal two-scale process has generator $\mathcal{L}_{\tilde{G}}$; that is, it has the law of the full-network process on $\tilde{G}$ started from the same configuration.*

Proof. Removals and within-community infections have identical rates in both processes. Consider a susceptible node $v \in V_j$ in configuration x. Under uniform placement, the two-scale rate at which v is infected by an import is

$$\sum_{i\neq j} \lambda_{ij}(x)\,\frac{1}{s_j(x)} = \sum_{i\neq j} \beta_{\mathrm{out}} W_{ij} \frac{y_i(x)}{N_i}\frac{s_j(x)}{N_j}\frac{1}{s_j(x)} = \sum_{i\neq j}\sum_{u\in V_i} \frac{\beta_{\mathrm{out}} W_{ij}}{N_i N_j}\mathbf{1}\{x_u = I\},$$

which is the total rate of the inter-community channels of $\tilde{G}$ that point into v. When $s_j(x) = 0$ both rates vanish. Hence all transition rates coincide and the generators are equal. □

Corollary 1. *If every inter-community bundle of G is already complete, $E_{ij} = V_i \times V_j$, with a common rate per pair of communities, then $\tilde{G} = G$ and the ideal two-scale process is exact.*

The surrogate preserves the total weight of every bundle, $\sum_{(u,v)\in V_i\times V_j} \tilde{\beta}_{uv} = \beta_{\mathrm{out}} W_{ij}$, but spreads it uniformly over all pairs (Figure 1c). The approximation error of the ideal method is therefore the difference between epidemics on G and on $\tilde{G}$, which differ only in their interfaces. The implemented method adds a numerical synchronisation error. Proposition 1 also provides a practical verification route: for small networks, the synchronisation error can be isolated by comparing the two-scale simulator with exact simulation on $\tilde{G}$.

4.2. Sources of discrepancy

Comparing G with $\tilde{G}$ gives four mechanisms by which the two-scale simulator can depart from the full-network process.

Boundary-state information (source side). In G, the rate of transmission from V_i to V_j is

$$a^{\star}_{ij}(x) = \beta_{\mathrm{out}} \sum_{(u,v)\in E_{ij}} w_{uv}\,\mathbf{1}\{x_u = I,\, x_v = S\}, \tag{7}$$

which depends on whether the particular bridge nodes are infected. In $\tilde{G}$, every infected node of V_i contributes. Exports can therefore occur before any bridge node is infected. Such *bridge bypass* is impossible in G and provides a direct diagnostic (Section 4.3).

Placement (receiver side). In G, a transmission infects the target endpoint of the firing channel; in $\tilde{G}$, the target is uniform among susceptible nodes. The location of an import determines its distance to the next interface and hence later crossings.

Interface exhaustion (receiver side). Under SI dynamics, a channel of G transmits at most once: afterwards its target is infected. A bundle with $|E_{ij}|$ channels therefore carries at most $|E_{ij}|$ transmissions, and a single-edge interface exactly one. In $\tilde{G}$, imports from V_i into V_j continue at rate $\lambda_{ij} = \beta_{\mathrm{out}} W_{ij} \bar{y}_i \bar{s}_j$ until V_j is exhausted; their expected number over a period $[t_1, t_2]$ is $\mathbb{E}\int_{t_1}^{t_2} \lambda_{ij}(s)\,ds$. When a receiving community remains largely susceptible for a long time, the two-scale simulator therefore keeps seeding it with additional infections that the full network cannot deliver.

Synchronisation. The midpoint quadrature and interpolation of Section 3.5 approximate (5).

Remark 1 (Placement under randomised interfaces). Suppose that the endpoints of each bundle are sampled uniformly and independently of the internal graphs and of the endpoints of other bundles, as in the experiments below. Then the target of the first transmission into a fully susceptible community is uniformly distributed in G, because it is the endpoint of a uniformly placed channel, and also in $\tilde{G}$. Placement can therefore affect arrival times only through transmissions after the first. In such designs, differences between the first crossing and later crossings isolate receiver-side effects (Section 6.6).

4.3. Single export: stochastic ordering and bridge bypass

Consider a source community V_0 of size n with an arbitrary internal graph, SI dynamics, and a single susceptible leaf node ℓ joined to a set $B \subset V_0$ of k bridge nodes by channels of rate β_{out}. Then $W_{0\ell} = k$, and before the export $\bar{s}_\ell = 1$, so that $\lambda_{0\ell} = \beta_{\text{out}} k\, \bar{y}_0$. Let τ_u be the infection time of node u, $A_u(t) = (t - \tau_u)^+$ its infection age, and $T_B = \min_{u\in B} \tau_u$.

Proposition 2 (Aggregate export clocks are early). *Let B be uniformly distributed over the k-subsets of V_0, independently of the internal graph, the initial configuration and the event randomness. Let T^{Full} and T^{TwoScale} be the first infection times of ℓ in the full network and in the ideal two-scale process. Then*

$$\mathbb{P}(T^{\text{Full}} > t) \;\geq\; \mathbb{P}(T^{\text{TwoScale}} > t) \qquad \text{for all } t \geq 0,$$

so that T^{TwoScale} is stochastically smaller than T^{Full} (Shaked and Shanthikumar, 2007). In particular, the mean and every quantile of the ideal two-scale export time are no larger than those of the full-network export time.

Proof. Before ℓ is infected, it does not influence V_0: it transmits nothing, and the rates of source nodes depend only on their infected neighbours. Both export times can therefore be constructed on one probability space from an autonomous SI trajectory X_0 of the source community, continued beyond the export, and an independent threshold $E \sim \text{Exp}(1)$:

$$T^{\text{Full}} = \inf\Big\{t : \beta_{\text{out}} \sum_{u\in B} A_u(t) \geq E\Big\}, \qquad T^{\text{TwoScale}} = \inf\Big\{t : \beta_{\text{out}} \frac{k}{n} \sum_{u\in V_0} A_u(t) \geq E\Big\},$$

where the second integrand follows from $\int_0^t \bar{y}_0(s)\, ds = n^{-1} \sum_u A_u(t)$. Conditional on X_0, the survival probabilities are $\exp(-\beta_{\text{out}} \sum_{u\in B} A_u(t))$ and $\exp(-\beta_{\text{out}} (k/n) \sum_u A_u(t))$. Since B is uniform and independent of X_0, $\mathbb{E}[\sum_{u\in B} A_u(t) \mid X_0] = (k/n) \sum_u A_u(t)$, and Jensen's inequality for the convex function $z \mapsto e^{-\beta_{\text{out}} z}$ gives $\mathbb{P}(T^{\text{Full}} > t \mid X_0) \geq \mathbb{P}(T^{\text{TwoScale}} > t \mid X_0)$. Taking expectations completes the proof. □

The inequality is strict whenever, with positive probability, the infection ages differ between nodes at time t and $0 < k < n$. It also holds when B is uniform among the nodes other than the initially infected node, because excluding the oldest infected node can only lower the conditional mean of the bridge hazard. The size of the gap is controlled by the conditional variance of the bridge hazard, for which sampling without replacement gives

$$\text{Var}\Big(\beta_{\text{out}} \sum_{u\in B} A_u(t) \;\Big|\; X_0\Big) = \beta_{\text{out}}^2\, k\, \frac{n-k}{n-1}\, \text{Var}_u\big(A_u(t)\big), \tag{8}$$

where Var_u is the population variance over source nodes. Once the source is fully infected, $\text{Var}_u(A_u(t)) = \text{Var}_u(\tau_u)$. The aggregate clock is therefore close to exact when node infection times are concentrated relative to $1/\beta_{\text{out}}$, as in a rapidly saturating clique, and is early when infection spreads slowly and unevenly through the source.

The same construction yields the probability of bridge bypass. In G, no export can precede T_B. In the ideal two-scale process,

$$\mathbb{P}\big(T^{\text{TwoScale}} < T_B\big) = \mathbb{E}\Big[1 - \exp\Big(-\beta_{\text{out}} \frac{k}{n} \int_0^{T_B} y_0(s)\, ds\Big)\Big], \tag{9}$$

Series	Structure	k	β	Runs per cell and model
Clique–leaf	source clique $n = 100$ + one leaf	2	0.005	10 000
Source–leaf sweep	source $n = 1000$, ASPL 1–6 in steps of 0.25, + one leaf	1, 5	0.003	400 (40 × 10)
Chain of cliques	$L = 10$ cliques of $n = 1000$	1	0.003	400 (40 × 10)
Chain size sweep	$L = 10$, $n \in \{50, 100, 250, 1000\}$, ASPL 1–6 in steps of 1	1, 5	0.003	400 (40 × 10)

Table 2: Experimental series. Each cell of a sweep uses 40 network realisations (blocks) with 10 interface and event replicates each. ASPL: target internal average shortest-path length (1 denotes the complete graph).

which is the probability that the aggregate hazard accumulated before the true interface becomes active triggers an export. It increases when the bridges are reached late and after many other nodes, which is the situation in sparse, long-path communities. Appendix A gives $\mathbb{E}[T_B]$ for a clique source.

4.4. A time-scale descriptor

Both source-side and receiver-side mechanisms act only while infection is still spreading inside a community. By (8), the source-side gap is governed by β_{out} times the dispersion of infection times. The expected number of surplus imports into a community during its internal spread is of order $\beta_{\text{out}} W_{ij}$ times the duration of that spread. Let σ_T denote the standard deviation of node infection times within a community, which measures this duration. We use the dimensionless ratio

$$\rho = \beta_{\text{out}}\, \sigma_T, \tag{10}$$

the internal timing spread in units of the mean single-edge crossing time $1/\beta_{\text{out}}$, as a descriptor of the expected accuracy. The two mechanisms scale differently with the interface width ($\sqrt{k}$ in (8) and k for surplus imports, partly offset because the full network itself admits up to k crossings). We therefore do not include k in ρ, and we test the descriptor empirically rather than derive an error law.

5. Experimental design

5.1. Benchmarks

Table 2 summarises the four experimental series. All use SI dynamics, unit contact weights, equal internal and external per-edge rates ($\beta_{\text{in}} = \beta_{\text{out}} = \beta$), and the scheduler of Algorithm 1 with $\tau = 1$. In the *clique–leaf* benchmark, a leaf is attached to $k = 2$ bridge nodes of a 100-node clique. The *source–leaf sweep* replaces the clique by connected random graphs of $n = 1000$ nodes with a prescribed average shortest-path length (ASPL). In the *chain* series, $L = 10$ communities are joined in a line by k edges between consecutive communities; infection starts in community 0. Connected random graphs are generated by adding uniformly random edges to a uniformly random labelled tree, with the number of added edges calibrated so that the ASPL lies within 0.1 of its target; ASPL 1 is the complete graph. Mean degree therefore decreases with the target ASPL. At a target ASPL of 6 it ranges from 2.08 for $n = 50$ to 3.4 for $n = 1000$. Interface endpoints are sampled uniformly, independently for each interface and replicate.[1] The initial infected node is pseudorandom and shared by both simulators, and it may be a bridge node; in the clique–leaf benchmark, the seed is a fixed node of the clique.

Full and TwoScale runs are matched: they share the graph, the interface and the initial node, but use independent event streams. All exports and arrivals entering the metrics were observed, so no censoring adjustment is needed. The horizon was 1000 for the clique–leaf benchmark and 30 000 otherwise. Two Full runs were extended to 40 000 to observe their terminal arrivals (Appendix C). The chain of cliques is the ASPL-1, $n = 1000$, $k = 1$ cell of the size sweep, reported separately because it is the most favourable case.

[1] In the source–leaf sweep, bridge sets are drawn from all nodes except one fixed label. Because the random graphs and the uniformly chosen seed are invariant in law under relabelling, this is equivalent to uniform placement, so the assumption of Proposition 2 holds.

5.2. Observables and metrics

For leaf benchmarks, we record the export time T_{export} (first infection of the leaf) and the bridge-activation time T_B. For chains, T_c is the first infection time of community c, measured from the initial infection; the terminal arrival is T_9. We report three summaries:

- the relative median shift $b_c = 100\,[\text{med}(T_c^{\text{TwoScale}}) - \text{med}(T_c^{\text{Full}})]/\text{med}(T_c^{\text{Full}})$, where negative values mean earlier arrival in TwoScale;
- the relative Wasserstein-1 distance $e_W = 100\,W_1/\overline{T}^{\text{Full}}$, where $W_1 = R^{-1}\sum_r |T_{(r)}^{\text{TwoScale}} - T_{(r)}^{\text{Full}}|$ is computed from the order statistics of the R times of each model;
- the bridge-bypass fraction, i.e. the percentage of TwoScale crossings that occur before any bridge node of the source community is infected.

For chains, the *local* shift of crossing c compares the mean of $T_c - T_{c-1}$ between models. We contrast the first crossing with the pooled crossings 2–9, using $T_9 - T_1$.

5.3. Statistical procedures

Confidence intervals are 95% percentile intervals from a bootstrap that resamples whole network blocks, keeping their ten replicates together, with 2000 resamples for chain summaries. Because e_W is positive even when two samples come from the same law, we calibrate it by a within-pair permutation test: the labels of the two runs in each matched pair are swapped with probability 1/2, and e_W is recomputed 2000 times. Under the null hypothesis that matched runs are exchangeable, this yields a sampling floor, which we summarise by its 95th percentile, and a p-value for the observed e_W. Appendix B gives an analytical approximation of the floor that applies when per-run data are not available.

The event seeds of a replicate depend on the target ASPL, the network block and the replicate index, but not on n or k; seeds are therefore shared across community sizes and interface widths (common random numbers). This reduces the variance of comparisons between such cells, but it also correlates their sampling errors. We therefore do not count agreement between cells as independent replication.

5.4. Verification of the reference

For a clique source with uniformly placed bridges, the expected bridge-activation time is known in closed form (Appendix A). Across the four clique-chain sizes and both interface widths, the mean T_B in Full agreed with the formula to within 2.3 standard errors. For $k = 1$, the mean waiting time after bridge activation (326.5–354.0) agreed with its exact value $1/\beta = 333.3$ to within sampling error (Table 3).

6. Results

6.1. Single export from a clique

Export times from a rapidly saturating source agree closely, but the aggregate interface admits impossible transmission orders. In the clique–leaf benchmark (Figure 2), mean export times were 110.18 (Full) and 112.62 (TwoScale); the matched difference of +2.43 had a bootstrap interval of $[-0.35, 5.27]$. The empirical CDFs nearly coincide ($W_1 = 3.25$, i.e. $e_W = 2.95\%$; Kolmogorov–Smirnov distance 0.0142, asymptotic $p \approx 0.27$). Of the mean export time of about 110, only $\mathbb{E}[T_B] = 8.36$ is needed on average to infect the first bridge node (11); the remainder is the wait for transmission across the interface, at a rate of at most $k\beta = 0.01$. The period in which boundary and community-wide states differ is therefore short, and the aggregate clock is close to the exact one, as Proposition 2 and (8) suggest.

The ordering diagnostic nevertheless separates the models: in 98 of 10 000 TwoScale runs (0.98%), the leaf was infected before either bridge node, which never happens in Full (Figure 2b,c). The middle panel shows $T_{\text{export}} - T_B$, a physical waiting time in Full but a signed ordering diagnostic in TwoScale.

6.2. Single export from connected random sources

As internal paths lengthen, bridge bypass becomes common and the aggregate export becomes early, as predicted by Proposition 2. Figure 3 shows the source–leaf sweep. Mean bridge activation times were similar in the two models, within 7% at every ASPL (e.g. 1165 and 1136 at ASPL 6, $k = 1$), as expected from identical

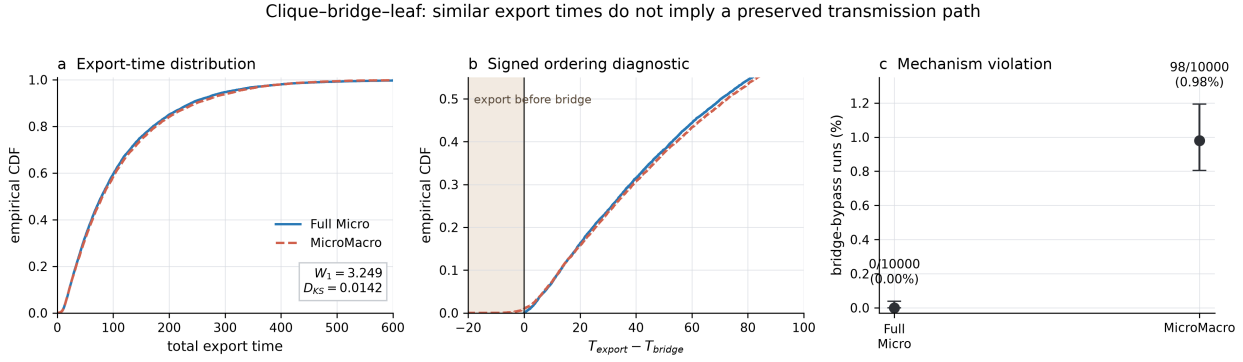


Figure 2: Clique–leaf benchmark ($n = 100$, $k = 2$, $\beta = 0.005$, 10 000 runs per model). (a) Empirical CDFs of the export time. (b) Empirical CDFs of $T_{\text{export}} - T_B$; negative values (shaded) are exports before any bridge node is infected. (c) Percentage of such bridge-bypass runs with 95% intervals. "Full Micro" denotes FULL and "MicroMacro" denotes TWOSCALE.

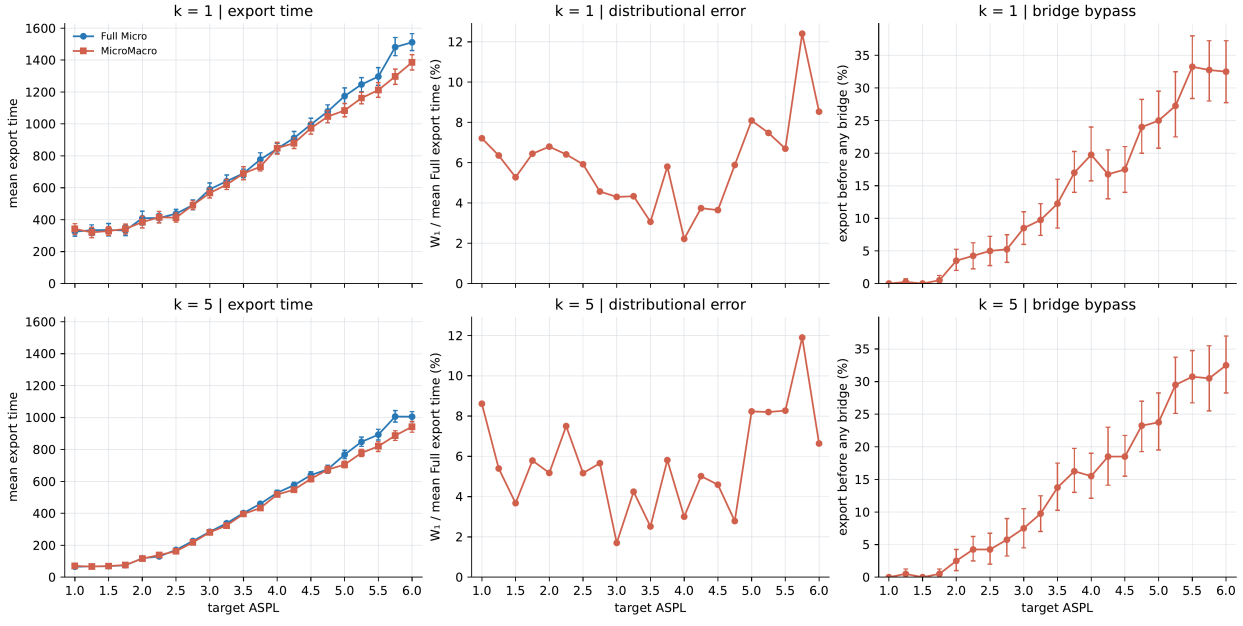


Figure 3: Source–leaf sweep: $n = 1000$, $\beta = 0.003$, $k = 1$ (top) and $k = 5$ (bottom), 400 matched runs per point. Left: mean export time. Centre: relative Wasserstein distance e_W of the export-time distributions. Right: percentage of TWOSCALE exports that precede the first bridge infection; FULL had none. Whiskers are 95% block-bootstrap intervals. "Full Micro" denotes FULL and "Production MicroMacro" denotes TWOSCALE.

local dynamics before the export. In FULL with $k = 1$, the wait after bridge activation stayed at 322–359 at every ASPL, consistent with Exp(β) with mean 333. In TWOSCALE, the signed interval $T_{\text{export}} - T_B$ ($k = 1$) shrank from about 340 in dense sources to 207–295 at ASPL ≥ 5, because aggregate hazard accumulated before the true bridge became infectious.

The bypass fraction rose from at most 0.5% for ASPL ≤ 1.75 to 19.75% (95% interval 15.8–24.0%) at ASPL 4 with $k = 1$, and to 32.5% for both widths at ASPL 6. The mean export shift was resolved, with intervals excluding zero, for both widths at every ASPL ≥ 5. Its values ranged from -6.2% to -12.4%; at

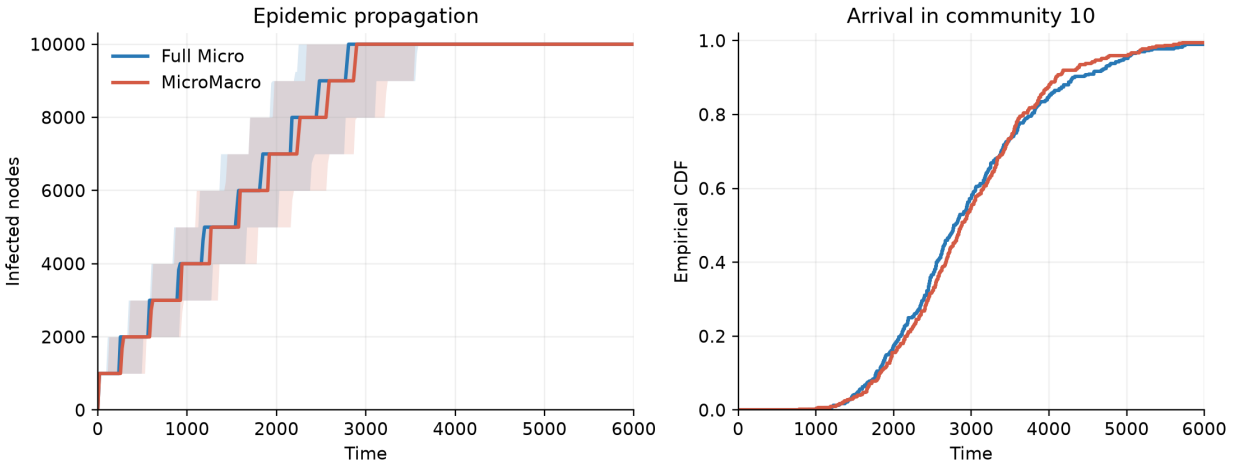


Figure 4: Chain of ten 1000-node cliques with one edge between consecutive cliques ($k = 1$, $\beta = 0.003$, 400 runs per model). Left: pointwise median number of infected nodes with interquartile bands. Right: empirical CDF of the arrival time in the tenth community (T_9). Four Full and two TwoScale arrivals lie beyond the displayed range; all were observed.

ASPL 6 they were -8.3% ($[-13.1, -3.5]$) for $k = 1$ and -6.2% ($[-10.3, -2.3]$) for $k = 5$. Below ASPL 5, a shift was resolved only for $k = 5$ at ASPL 3.75, 4.25 and 4.5 (between -3.7% and -5.8%). No positive shift was resolved anywhere in the sweep, in agreement with the direction predicted by Proposition 2.

The irregular centre panel requires care. For nearly exponential export times and $R = 400$ runs per model, the expected e_W of two samples from the same law is about $\sqrt{\pi/R} \approx 9\%$ (Appendix B). The values of 3.7–8.6% observed for ASPL ≤ 1.75 are therefore consistent with pure sampling variability. Single-export distances should thus be read together with the mean shift and its interval. Bypass, by contrast, is zero in Full by construction, so any observed bypass is a genuine discrepancy. At ASPL 4 it affects one fifth of the exports while the export-time distributions remain close.

6.3. Repeated crossings through a chain of cliques

When internal spread is fast, close agreement survives nine successive aggregate crossings. In the chain of ten 1000-node cliques with $k = 1$, the median terminal arrival was 2785 in Full and 2877 in TwoScale, and $e_W = 3.34\%$ (Figure 4). The permutation floor of this cell was 5.1% ($p = 0.32$), so the two terminal distributions are statistically indistinguishable. The CDFs cross, so the difference is not a fixed delay. A 1000-node clique saturates on average within $2H_{999}/(\beta n) \approx 5$ time units, compared with a mean edge-crossing time of 333 ($\rho = 0.002$). The aggregate rate therefore differs from the exact one only during a negligible fraction of each crossing.

6.4. Chains of random communities

Sparse, long-path communities produce systematically early arrivals whose relative size is stable along the chain while the absolute lead accumulates. Figure 5 shows the cumulative arrival times in chains of 1000-node communities. For target ASPL 2–3, terminal e_W was 1.4–2.5%, close to its sampling floor. At ASPL 6, TwoScale reached the last community earlier by 13.9% in median for $k = 1$ and by 10.7% for $k = 5$ ($e_W = 13.6\%$ and 10.9%). Along the ASPL-5 row with $k = 1$, the relative median lead was about 10% at both the first and the last community, while its absolute size grew from about 107 to 1018 time units. A constant relative error along a chain therefore means a growing absolute error at distant communities. The positive first-column values in the clique row (+12.6% and +11.5%) are not resolved: for $k = 1$, the first-crossing mean shift is +4.9% with a 95% interval of $[-8.0, 18.3]\%$. Single-crossing distances also have a larger sampling floor than cumulative ones.

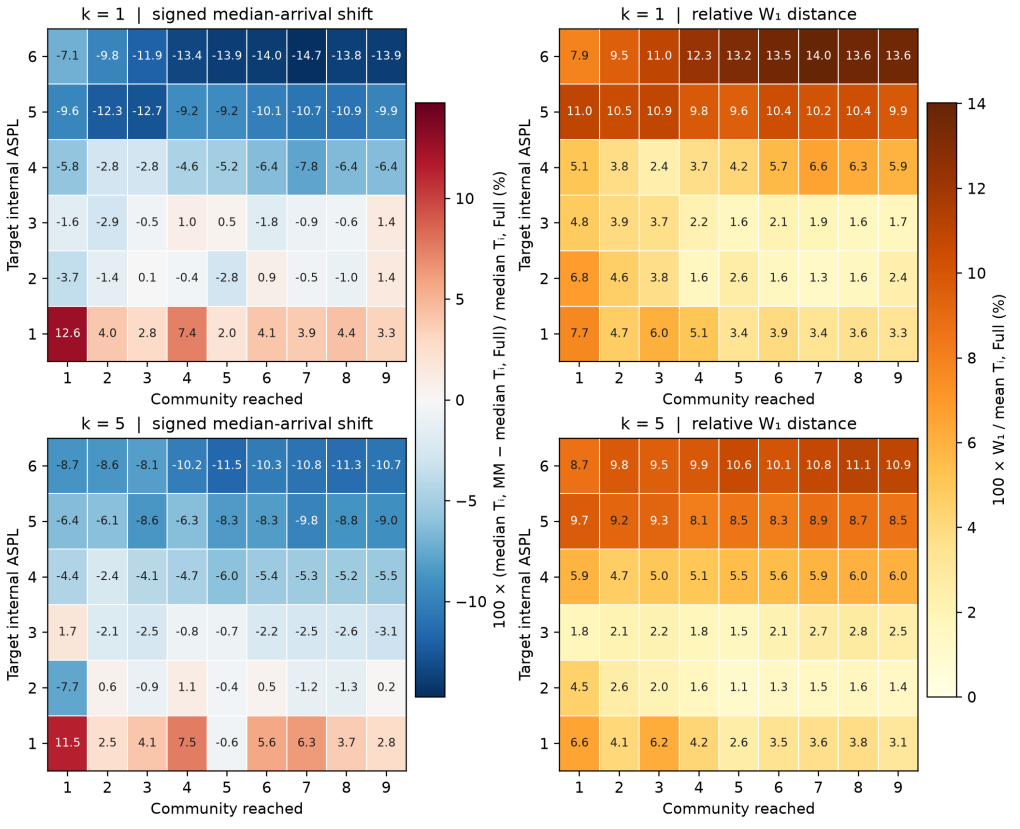


Figure 5: Cumulative arrival accuracy in ten-community chains with $n = 1000$, by target ASPL (rows) and community reached ($c = 1, \ldots, 9$, columns), for $k = 1$ (top) and $k = 5$ (bottom). Left: relative median shift b_c (negative: TwoScale earlier). Right: relative Wasserstein distance. Each cell uses 400 matched runs; no uncertainty is shown (see Figure 6 for the terminal arrivals).

6.5. Community size and interface width

The same internal path length is much less favourable in small communities. Figure 6 reports the terminal discrepancy for all 48 configurations of the size sweep. The permutation floor ranged from 1.6% to 5.9%. In the 16 configurations with ASPL ≤ 2, e_W was 1.1–4.4%, and in 14 of them it was not distinguishable from the floor at the 5% level. In 31 of the 32 configurations with ASPL ≥ 3, e_W exceeded the floor ($p < 0.01$), and the median shift was negative with a block-bootstrap interval excluding zero. The exception was $n = 1000$, ASPL 3, $k = 1$, with $e_W = 1.7\%$. At a fixed ASPL ≥ 3, the discrepancy decreased monotonically with community size for both interface widths. At ASPL 6, e_W fell from 41.3% for $n = 50$ to 13.6% for $n = 1000$ with $k = 1$, and from 32.4% to 10.9% with $k = 5$. Wider interfaces reduced the discrepancy in sparse communities but did not remove it. The fraction of TwoScale crossings that preceded infection of any bridge node of the source community rose from 0.1–2.3% at ASPL 1 to 30–52% at ASPL 6.

6.6. A single time-scale ratio organises the discrepancy

Expressed through $\rho = \beta\sigma_T$, the results of all sizes, structures and interface widths collapse onto one curve. We computed σ_T in each configuration as the standard deviation of node infection times within each source community of the Full runs, averaged over communities and runs. Figure 7(a,b) plots the terminal e_W and b_9 against ρ. The rank correlation between e_W and ρ was 0.92 over all 48 configurations and 0.98 over the 32 configurations with ASPL ≥ 3; the corresponding values for target ASPL were 0.82 and 0.67. In the 18 configurations with $\rho < 0.25$, $e_W \leq 4.4\%$, and 15 were indistinguishable from the permutation floor. All 30 configurations with $\rho \geq 0.25$ were resolved. For $\rho \geq 0.3$, a log–log fit gave $e_W \approx 15\rho\,\%$ (slope 1.03; median relative deviation 8%, maximum 41%), and $b_9 \approx -e_W$, i.e. the discrepancy was almost entirely an



Figure 6: Terminal-arrival accuracy in ten-community chains for all 48 configurations of the size sweep. Colour and bold numbers: relative Wasserstein distance e_W of T_9 (%). Italic numbers: relative median shift b_9 (%; negative means TwoScale earlier). Hatched cells: e_W not distinguishable from its within-pair permutation floor ($p \geq 0.05$). Each cell: 400 matched runs.

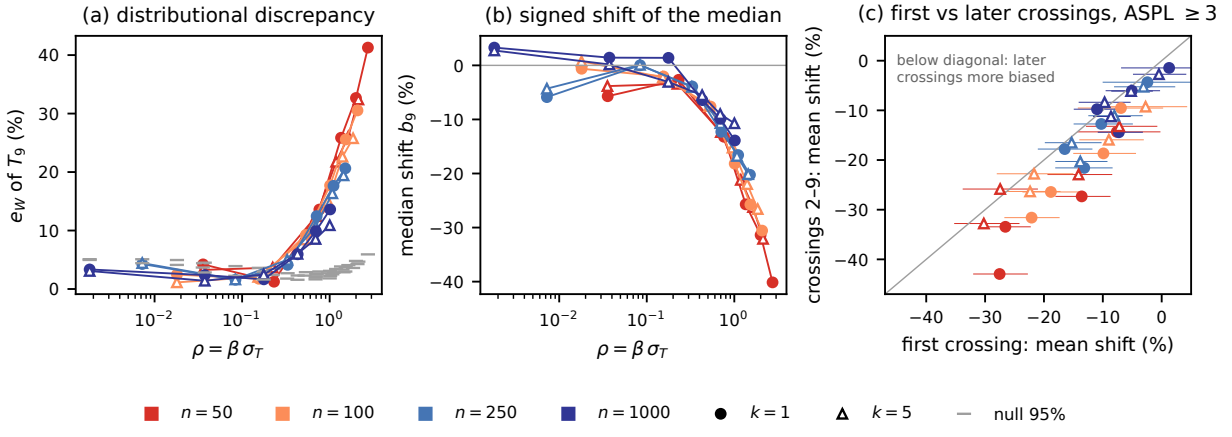


Figure 7: (a) Relative Wasserstein distance and (b) relative median shift of the terminal arrival against $\rho = \beta\sigma_T$, for all 48 chain configurations (colours: community size; filled circles: $k = 1$; open triangles: $k = 5$; lines connect ASPL 1–6 at fixed n and k). Grey ticks in (a): 95th percentile of the within-pair permutation floor of each configuration. (c) Local mean shift of the first crossing against that of crossings 2–9, for configurations with ASPL ≥ 3, with 95% block-bootstrap intervals of the first-crossing shift.

early shift. We also examined two alternative scalings: $\beta\sqrt{k}\sigma_T$, suggested by (8), and the mean time from a community's first infection to infection of its first bridge node, multiplied by βk. Both separated the two interface widths more than ρ did.

Figure 7(c) separates the crossings. By Remark 1, the first crossing in our design is subject only to source-side and numerical mechanisms, whereas later crossings are also subject to placement after the first

import and to surplus imports. For $k = 1$, later crossings were more strongly biased than the first in 15 of the 16 configurations with ASPL ≥ 3. The difference was resolved in 9 of them, including all four sizes at ASPL 6; for example, -27.5% for the first crossing against -42.9% for crossings 2–9 at $n = 50$. For $k = 5$, the difference was smaller (14 of 16 point estimates negative, 3 resolved). This dependence on k is what interface exhaustion predicts: a single physical edge permits one crossing, whereas five edges permit up to five, so surplus aggregate imports matter less. Because seeds are shared between configurations, we regard this pattern as supporting rather than conclusive evidence. A direct count of surplus imports would require event-type logs that were not stored.

7. Discussion

7.1. Practical guidance

The results suggest a simple procedure for deciding whether aggregate coupling is adequate.

Decide which observable matters. For the timing of spread between communities, the two-scale simulator can be accurate. For transmission routes, it cannot be trusted. Up to half of its crossings in sparse communities, and a fifth of exports even where export-time distributions agreed, occurred before any true bridge node was infected. Questions that depend on bridge nodes, such as targeting interventions at the individuals who connect communities or attributing an outbreak to a source, require boundary information that the aggregate interface does not contain.

Estimate the time-scale ratio. Compute $\rho = \beta_{\text{out}}\sigma_T$. Since σ_T describes spread within a single community, it can in principle be estimated from the local engines alone, by simulating an isolated community from a uniformly placed seed. We computed it from the reference runs and have not tested this local estimate. In our benchmarks, terminal arrivals were statistically indistinguishable for $\rho \lesssim 0.25$. For $0.3 \lesssim \rho \lesssim 3$, TwoScale arrived early by roughly $15\rho\%$. This rule is an empirical summary of one family of synthetic networks and parameters, not a general error law.

Check absolute as well as relative errors. Along a chain, the relative discrepancy was nearly constant, so the absolute lead at distant communities grew in proportion to the travel time. For intervention timing at a specific destination, the absolute discrepancy at that destination is the relevant quantity.

Calibrate distances against sampling variability. With a few hundred runs per model, distances of a few percent between arrival-time distributions are expected even between identical models (Appendix B). Permutation baselines or analytical floors should accompany such comparisons.

7.2. Computational cost

The benchmarks were designed for accuracy assessment, so their explicit interfaces contain at most five edges and the full reference is cheap. In this regime, aggregation removes almost no work, and the cost of grid synchronisation dominates. The median wall-clock time per replicate of the Python implementations ranged from 0.04–0.07 s (Full) and 0.6–31 s (TwoScale) for $n = 50$ to 8.6–18 s and 6.7–124 s for $n = 1000$. For $n = 1000$, the ratio of the two medians ranged from 0.76 (ASPL 2, $k = 5$) to 6.9 (ASPL 6, $k = 1$). Two conclusions follow. First, in settings where the full network is known and small, Full should be preferred. Second, the overhead of the two-scale method stems from the black-box scheduler rather than from aggregation. An event-driven scheduler that integrates the piecewise-constant hazard exactly (Section 3.5) would remove both this overhead and the synchronisation error. Computational gains are expected when inter-community contacts are numerous or unavailable, which is the setting the method is intended for (Kuryliak and Emmerich, 2025); quantifying them is left to future work.

7.3. Limitations

The experiments use SI dynamics. With recovery, bridge nodes can stop transmitting while the community-wide fraction remains positive, and the aggregate rate may then also overestimate exports late in a local outbreak. All networks are synthetic, the macro-topology is a chain, and internal and external per-edge rates are equal. Interfaces were placed at random, which by Remark 1 neutralises placement error for

first imports. Structured interfaces, for example with bridges at hubs or with incoming and outgoing bridges far apart, may behave differently. The synchronisation step was not varied, so the numerical component of the discrepancy was not isolated; Proposition 1 provides a direct way to do so. Finally, ρ was evaluated with reference runs and chosen among a small number of candidate scalings; its predictive value for other network families remains to be tested.

7.4. Possible extensions

The analysis points to modifications that use only aggregate information. Interface exhaustion can be represented by limiting the number of imports per bundle to its number of channels, which is known whenever W_{ij} counts edges, or by reducing λ_{ij} as this capacity is consumed. When partial boundary information is available, $\bar{y}_i$ in (4) can be replaced by the infected fraction among known bridge nodes. Placement can then be restricted to known receiving endpoints. Each such variant can be evaluated against FULL and against exact simulation on $\tilde{G}$, which separates modelling from numerical error.

8. Conclusions

The two-scale method couples exact local Gillespie engines through aggregate inter-community hazards. In its ideal form, it is exact simulation on a surrogate network in which every interface is smeared uniformly over all pairs of nodes. This identifies what the method loses: the state of the actual bridge nodes, the location of imports and the finite capacity of thin interfaces. For randomly placed interfaces, it also implies that aggregate export clocks are stochastically early. In controlled comparisons with full-network Gillespie simulation, terminal arrival-time distributions were statistically indistinguishable when infection spread through communities quickly relative to the time needed to cross an interface edge. As the internal timing spread approached and exceeded that crossing time, the two-scale simulator arrived systematically early, by up to 40% in small sparse communities. The ratio $\rho = \beta\sigma_T$ organised these results across community sizes, internal structures and interface widths. Close timing never guaranteed admissible transmission routes. The method is therefore a reasonable tool for propagation-time questions when ρ is small. It is not suited to questions about individual transmission routes, or to sparse communities with long internal paths unless the interface is refined.

A. Clique formulas and verification

Consider an n-node clique with SI dynamics, per-edge rate β, a uniformly placed seed and a uniformly placed set of k bridge nodes. When m nodes are infected, the next internal infection occurs at rate $\beta m(n-m)$, independently of which nodes are infected, and the infected set is a uniform m-subset. The probability that none of the first m infected nodes is a bridge is $\binom{n-k}{m}/\binom{n}{m}$, hence

$$\mathbb{E}[T_B] = \sum_{m=1}^{n-k} \frac{\binom{n-k}{m}}{\binom{n}{m}} \frac{1}{\beta m(n-m)}, \qquad \mathbb{E}[T_B] = \frac{H_{n-1}}{\beta n} \quad (k=1), \tag{11}$$

where H_{n-1} is the harmonic number; the case of a bridge seed ($T_B = 0$) is included. For $k = 1$, the export time in FULL is T_B plus an independent $\mathrm{Exp}(\beta)$ waiting time. The expected time to infect the whole clique is $2H_{n-1}/(\beta n)$. Table 3 compares (11) with the first community of the clique chains.

B. Sampling floor of the Wasserstein distance

Let F_R and G_R be empirical CDFs of two independent samples of size R from the same continuous law F. Then $W_1 = \int |F_R - G_R|\,dx$, and $\sqrt{R}\,(F_R - G_R)$ is approximately $\sqrt{2}$ times a Brownian bridge evaluated at $F(x)$ (del Barrio et al., 1999). Since $\mathbb{E}|B(u)| = \sqrt{2u(1-u)/\pi}$ for a Brownian bridge B,

$$\mathbb{E}[W_1] \approx \frac{2}{\sqrt{\pi R}} \int \sqrt{F(x)\big(1-F(x)\big)}\,dx.$$

For an exponential law with mean μ, the integral equals $\mu\pi/2$, so that $\mathbb{E}[W_1]/\mu \approx \sqrt{\pi/R}$, which is 8.9% for $R = 400$ and 1.8% for $R = 10\,000$. Arrival times that are sums of several crossings are less dispersed relative

n	k	$\mathbb{E}[T_B]$, Eq. (11)	FULL mean T_B	FULL wait after T_B
50	1	29.86	29.72 ± 0.71	354.0 ± 16.9
100	1	17.26	17.05 ± 0.38	326.5 ± 17.0
250	1	8.13	7.84 ± 0.15	347.4 ± 16.6
1000	1	2.49	2.51 ± 0.04	353.8 ± 17.9
50	5	15.97	15.96 ± 0.55	–
100	5	10.31	9.64 ± 0.29	–
250	5	5.35	5.16 ± 0.12	–
1000	5	1.80	1.77 ± 0.03	–

Table 3: Verification of the reference simulator on the first community of the clique chains ($\beta = 0.003$, 400 runs per cell): mean bridge-activation time and, for $k = 1$, mean wait after activation (exact value $1/\beta = 333.3$). Values are mean ± standard error.

to their mean, and their floor is correspondingly lower; the permutation floors of the chain configurations ranged from 1.6% to 5.9%.

C. Reproducibility details

Seeds are derived deterministically from the experiment parameters. Each TWOSCALE engine draws from a private pseudorandom stream; midpoint probes copy the stream state of their engine and do not advance it. Two FULL runs ($n = 50$, ASPL 6, $k = 1$) had not reached the tenth community by $T = 30\,000$. They were rerun to $T = 40\,000$ with the same graphs, initial nodes and event seeds. Their event histories up to 30 000 were verified to match the saved runs, and their terminal arrivals occurred at about 31 059 and 31 007. No horizon value was substituted for an unobserved event. The permutation and bootstrap analyses of the chain configurations, the descriptor σ_T and Figures 6–7 were computed from the saved runs by the scripts `extrascripts/analysis/smpt_twoscale_paper_metrics.py` and `extrascripts/visualization/smpt_twoscale_paper_figures.py`.

CRediT authorship contribution statement

Yulian Kuryliak: Conceptualization, Methodology, Software, Investigation, Formal analysis, Visualization, Writing – original draft. **Michael T. M. Emmerich:** Conceptualization, Methodology, Supervision, Writing – review & editing. [Authors to confirm.]

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The simulation code, saved simulation outputs and analysis scripts are available from the corresponding author on reasonable request. [Replace with a repository DOI if the data are deposited.]

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used [name of tool or service] in order to [reason]. After using this tool or service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. [Authors to complete.]

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